

Probability and Statistics (MT-2005)

Homework # 02(c)

Basic Probability Theory

1. Using definition of the classical probability

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes in the sample space}}$$

and set theory, prove that:

$$P(A \setminus B) = P(A) - P(A \cap B)$$

Discuss what would be the result if $B \subseteq A$. **Hint:** Use the fact that $A = (A \cap B) \cup (A \setminus B)$ and apply the classical probability formula.

2. Prove that if A and B are independent, then their complements A^c and B^c are also independent, i.e.,

$$P(A^c \cap B^c) = P(A^c)P(B^c)$$

Hint: Use the identity $A^c \cap B^c = (A \cup B)^c$ and express everything in terms of $P(A)$ and $P(B)$.

3. Prove that if $A \subseteq B$, then:

$$P(A \cap B) = P(A)$$

Hint: Use the definition of a subset and the fact that the intersection of a set with a superset is the smaller set.

4. For three events A, B, C , prove that:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

5. The definition of conditional probability states:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Prove that:

$$P(A^c|B) = 1 - P(A|B)$$

Hint: Use the fact that $A^c \cup A = S$ and take the probability of both sides.